

Takrim Nahin

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EDUCATION

University of Central Florida

Orlando, FL

B.S. Computer Science | GPA: 3.71

Expected December 2026

- **Relevant Coursework:** Data Structures & Algorithms, Computer Architecture, Algorithms for Machine Learning, Systems Software, Processes for Object-Oriented Software Development

EXPERIENCE

Technical Systems & Automation Intern

Apr. 2025 – Jul. 2025

Beige Corporation (Automation & Workflow Solutions)

Remote

- Collaborated with cross-functional teams across Sales, Production, and Post-Production to integrate AI-assisted workflow automations in Zapier and Make.com, eliminating 10+ manual handoffs and supporting enterprise-wide AI adoption across 4 business units
- Engineered agentic AI workflows on OpenAI API and Google Workspace APIs, designing and iterating structured prompts that reduced errors across 3 production processes through repeatable AI-driven data routing and decision logic
- Documented AI workflow logic, prompt assumptions, and API data flows across 4 production integrations, capturing model behavior and edge cases to support long-term maintainability and responsible deployment

PROJECTS

SyncedStudy – Real-Time Collaborative Study Platform | React, Node.js, Express, Socket.io

June 2026

- Architected a real-time collaborative Pomodoro platform with server-authoritative timer state broadcast via Socket.io, eliminating client drift across distributed sessions and maintaining consistent state for up to 6 concurrent users per room
- Engineered a full room lifecycle system — random code generation, automatic host role delegation on disconnect, empty-room cleanup, and server-side message rate limiting — supporting up to 30 concurrent active rooms with no database dependency
- Deployed full-stack application across Vercel (React) and Railway (Node.js/Express) with a live public endpoint; implemented a configurable Pomodoro state machine, per-user focus time tracking, and full task history timeline in the session summary

DocuQuery – AI-Powered Document Q&A | Python, Flask, OpenAI API

May 2026

- Engineered and deployed a production RAG application enabling users to upload any PDF and query its content via natural language; shipped via a Git-based CI/CD pipeline to Render's cloud platform with automated builds on every push
- Implemented an end-to-end ML pipeline: PDF text extraction and preprocessing with PyMuPDF, semantic chunking, cosine similarity retrieval over OpenAI text-embedding-3-small embeddings, and GPT-3.5-turbo answer generation
- Built for non-technical end users, enabling plain-language document queries through a simple upload interface with no setup required; deployed to a live public endpoint on Render with full end-to-end ownership from development through release

IRAD Budget & Asset Tracker | Python, FastAPI, SQLAlchemy, SQLite

April 2026

- Engineered a full-stack budget and asset management system using FastAPI and SQLAlchemy, exposing 15 documented REST endpoints to track project budgets, equipment assignments, and division-wide status with SQLite persistence
- Implemented over-budget detection at line-item and project levels with computed analytics (utilization %, remaining funds, portfolio value); built an interactive web dashboard with automated CSV export, Swagger UI documentation, and a schedulable CLI for hands-free weekly reporting

RESEARCH & PUBLICATIONS

Quantum Neural Networks in Machine Learning

April 2026

- Achieved 96% test accuracy on digit classification using only 24 trainable parameters by designing a novel white/black amplitude encoding scheme in Qiskit – outperforming classical equivalents in parameter efficiency
- Co-authored a research paper on QNN architecture, training and validation methodology, and real-world applications spanning image recognition, drug discovery, and financial prediction

TECHNICAL SKILLS

Languages: Python, Java, C, SQL, HTML, CSS, JavaScript

Frameworks & Libraries: React, Node.js, Express, Socket.io, FastAPI, Flask, SQLAlchemy, Qiskit, NumPy, PyMuPDF

Tools & Platforms: Git, Zapier, Make.com, OpenAI API, Google Workspace, Render, Vercel, Railway, DigitalOcean

Concepts: Machine Learning, Prompt Engineering, Data Preprocessing, Model Evaluation, REST APIs, Retrieval-Augmented Generation, Real-Time Systems, Agile Development, CI/CD Pipelines